Hierarchical Clustering Numerical.

4 Samples.

A	1
B	2
C	3
D	8

For ease : Make it as Matrix.

	A	B	C	D
A	0	1	4	7
B	1	0	3	6
C	4	3	0	3
D	7	6	3	0

Use complete linkage

use maximum distance

*. Merge points with most similar.

{A, B} , {C}, {D}

A B distance least
thence most similar

Update Matrix

	{A, B}	{C}	{D}
{A, B}	0	4	7
{C}	4	0	3
{D}	7	3	0

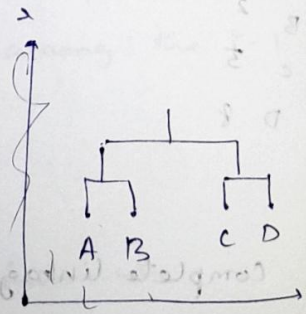
Here comes linkage

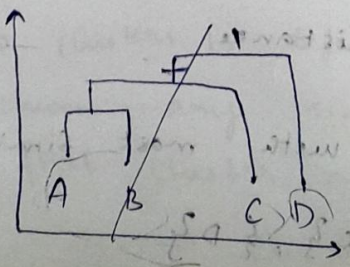
$d(A, C) = 4$
$d(B, C) = 3$ } $d(\{A, B\}, C) = 4$

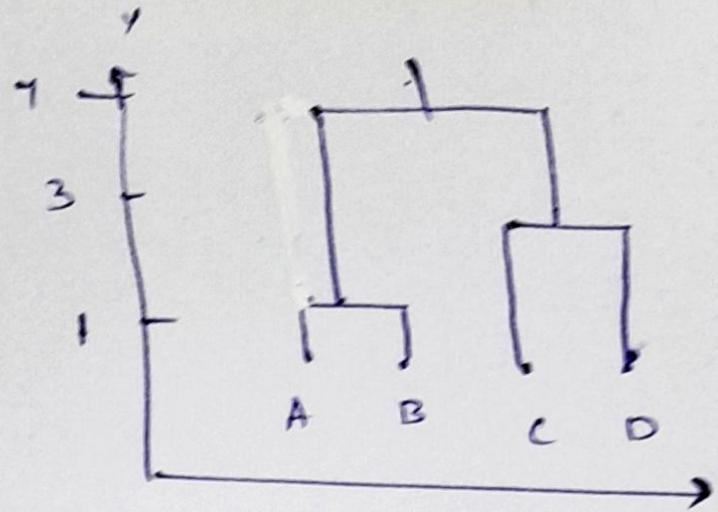
$d(A, D) = 7$
$d(B, D) = 6$ } $d(\{A, B\}, D) = 7$

Now clustering C & D as it is minimum

	{A, B}	{C, D}
{A, B}	0	7
{C, D}	7	0







you can cut anywhere.